

Paper 2 — Dynamic Factor Allocation via Regime-Switching Signals (Shu & Mulvey — 2024)

"Is this factor currently in a bull or bear regime — i.e., is it likely to keep out- or under-performing the market over the coming weeks?"

That's it. For each factor it gives you a current-state label:

- Value → bull or bear
- Size → bull or bear
- Momentum → bull or bear
- Quality → bull or bear
- Low-vol → bull or bear
- Growth → bull or bear

RAW DATA TO COLLECT (11 series):

Bloomberg (daily total return, 1993–present):

- [] Market — MSCI USA Index
- [] Value — MSCI USA Enhanced Value + Value Weighted (splice 1997/12)
- [] Size — MSCI USA Low Size + Equal Weighted (splice 1994/6)
- [] Momentum — MSCI USA Momentum SR Variant + Momentum (splice 2002/6)
- [] Quality — MSCI USA Sector Neutral Quality + Quality (splice 1998/12)
- [] Low Vol — MSCI USA Min Volatility + Risk Weighted (splice 1994/6)
- [] Growth — Russell 1000 Growth Index

FRED (daily, free):

- [] VIXCLS — VIX
- [] DGS2 — 2Y Treasury yield
- [] DGS10 — 10Y Treasury yield
- [] DGS3MO — 3M Treasury yield (risk-free)

ALSO NEEDED (not data — tools/params):

- [] jumpmodels library (GitHub, free)
- [] Transaction cost assumption: 5 bps each side
- [] BL params: risk aversion $\delta=2.5$, covariance EWMA halflife=126d

Most of them are available in Bloomberg.

Execution:

Stage 1 — collect raw data

7 index series (Bloomberg)
Market + 6 factors, daily, 1993→now

4 macro series (FRED)
VIX, 2Y, 10Y, 3M yields

Stage 2 — transform

Active returns
factor return – market return

Aligned macro inputs
log-diffs, smoothing

Stage 3 — feature engineering

~20 features per factor
EWMA returns, RSI, %K, MACD, downside dev, beta + macro

Stage 4 — the model (run once per factor)

Sparse Jump Model
clusters periods into 2 states + jump penalty for stability

Stage 5 — the signal

Bull / bear label per factor (today)
6 signals: value, size, momentum, quality, low-vol, growth

Stage 6 — allocate

Black-Litterman
turns signals into views

Portfolio weights
long-only, fully invested, 7 assets

The SJM is publicly available along with the Black-litterman.